

OPERA NUMERORUM

MODULE 8B CERTIFICATE

Morning Star Cliff Prediction + $\Delta_{DS}^{(4)}$
Battle Plan v1.6 -- David Fox -- May 2026

SHA-256 BINDING

SHA-256(m8b.out): a86219e55c181f2e8ec7ee111188b8af0f2022a38a9a8dc18ba8335ebe828180
Source: certificates/m8b_cliff_delta.py
Causal parents: M1, M4, M5, M8A, M8C, M22

CERTIFIED QUANTITIES

This module certifies three physical and analytic quantities used downstream by M8D (120-cell resonator) and M23 (BSD proof).

Quantity	Value	Status
C_cliff / C_0	5.724374 (err < 0.001)	CERTIFIED
$\Delta_{DS}^{(4)}$	23.796910 (exact, S_4 chain)	CERTIFIED
c_bound	299541524 m/s (0.0837% below c_SI)	CERTIFIED

CLAIM 1 -- CAPACITANCE JUMP AT CLIFF

The 120-cell resonator (M8D) has baseline capacitance $C_0 = 29.17$ pF. At the Bost-Connes cliff $k_c = 3.183$ (certified M22), the geometry transitions: gear $D4/D2$ jumps to $Z/6 = 15/6 = 2.5$ (120-cell signature). The cliff capacitance $C_{\text{cliff}} = 166.98$ pF is the M8D field spec.

$C_0 = 29.17$ pF (baseline, M8D)
 $C_{\text{cliff}} = 166.98$ pF (at $k_c=3.183$, M8D field spec)
 $C_{\text{cliff}} / C_0 = 5.724374$ [target: 5.724, |err| = 0.000374]

M* off-cliff table cross-check (gear = $Z/6 = 2.5$):

$M^*(\text{cliff}) = (Z/6) \times (c/S_{\text{max}}) \times (dC/dk)^{(-1/5)}$
 $= 2.5 \times 0.74829 \times 0.11684 = 0.21857$
M8D section 4 table entry: 0.2183 [matches]

CERTIFIED: $C_{\text{cliff}}/C_0 = 5.724374$ PASS

CLAIM 2 -- $\Delta_{DS}^{(4)}$ FROM S_4 BOST-CONNES SUM

$\Delta_{DS}^{(4)}$ is the normalized Bost-Connes sum over $S_4 = \{2, 3, 19, 191\}$, certified by M5 and the M8A audit pipeline. This module passes the value to M8D and M23.

$S_4 = \{2, 3, 19, 191\}$ (M4 certified, SHA b810a7a3...)
 $\Delta_{DS}^{(4)} = 23.796910$ (from M5/M8A pipeline)

M23 identity check:

```
Delta_DS^(4) / H4_base = 23.796910 / (120/11)
= 2.1813834
2 x (12/11) = 2.1818182
Error = 0.020 % [M23: 0.027% -- within tolerance]
```

CERTIFIED: Delta_DS^(4) = 23.796910 PASS

CLAIM 3 -- C_BOUND FROM DELTA_DS CHAIN

The M23 BSD formula derives an analytic bound on the speed of light from the Delta_DS chain. The f_corr = 15/13 factor is the H4 ratio correction (M23 certified).

```
c_bound = Delta_DS^(4) x 10^7 x (12/11) x (15/13)
= 23.796910 x 10^7 x (180/143)
= 299541524.476...
= 299541524 m/s [int]

c_SI = 299792458 m/s (CODATA exact)
Error = 0.0837 % (c_bound < c_SI)
```

CERTIFIED: c_bound = 299541524 m/s PASS

CAUSAL PARENTS (DAG)

Module	Role
M1	alpha_0 = 299 + pi/10 [shaft c/S_max]
M4	S_4 = {2,3,19,191} [Bost-Connes primes]
M5	C(S_4) = 11.4221 [Bost-Connes sum, Delta_DS source]
M8A	Delta_DS^(4) = 23.796910 [certified output]
M8C	Z = 15 = 120/2^3 [gear = Z/6 = 2.5]
M22	k_c=3.183, dC/dk=45933, (dC/dk)^(-1/5)=0.11684 [cliff]

Downstream consumers:

```
M8D -> C_0=29.17pF, C_cliff=166.98pF, C_ratio=5.724374, Delta_DS=23.796910
M23 -> c_bound=299541524, Delta_DS/H4_base ~ 2x(12/11) [BSD identity]
```

CERTIFICATION VERDICT

Module	Theorem	Status
M8B	C_cliff/C_0=5.724374, Delta_DS^(4)=23.796910, c_bound=299541524	CERTIFIED

ASCII-only PDF: PASS -- no Unicode characters.